

Maximum likelihood estimation

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Recap

Definition: The *likelihood* $L(Model) = P(Data|Model)$ of a model is the probability of the observed data, given that we assume a certain model and certain values for the parameters that define that model.

Coin example: flip a coin 5 times, with $\pi = P(Heads)$

- + Model: $Y_i \sim \text{Bernoulli}(\pi)$, and $\hat{\pi} = 0.9$
- + Data: $y_1, \dots, y_5 = T, T, T, T, H$
- + Likelihood: $L(0.9) = P(y_1, \dots, y_5 | \pi = 0.9) = 0.00009$

Recap

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\pi) = (1 - \pi)^4(\pi)$
- + Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

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How do I maximize functions?

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

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- + Observed data: T, T, T, T, H
- + Likelihood: $L(\pi) = (1 - \pi)^4(\pi)$
- + Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

Step 1: Write down the likelihood

$$L(\pi) = (1 - \pi)^4(\pi)$$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

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- + Observed data: T, T, T, T, H
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- + Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

Step 2: Take the log

$$\log L(\pi) =$$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

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- + Observed data: T, T, T, T, H
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Step 3: maximize

$$\log L(\pi) = 4 \log(1 - \pi) + \log(\pi)$$

How do we find where maxima/minima occur for a function?

Maximum likelihood estimation with calculus

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\pi) = (1 - \pi)^4(\pi)$
- + Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

Step 3: maximize -- take first derivative and set equal to 0

$$\log L(\pi) = 4 \log(1 - \pi) + \log(\pi)$$

$$\frac{d}{d\pi} \log L(\pi) =$$

Class activity

Work on the activity (handout) with a neighbor, then we will discuss as a class

Class activity

+ $P(Y_i = 0) = \pi_0$

+ $P(Y_i = -1) = 2\pi_0$

+ $P(Y_i = 1) = 1 - 3\pi_0$

Observe data $-1, -1, 0, 1, 0, -1$.

$$L(\pi_0) = ?$$

Class activity

$$\log L(\pi_0) = 3 \log(2) + 3 \log(\pi_0) + 2 \log(\pi_0) + \log(1 - 3\pi_0)$$

$$\frac{d}{d\pi_0} \log L(\pi_0) =$$

Maximum likelihood in more generality

Suppose that $Y_i \sim \text{Bernoulli}(\pi)$. We observe n observations $Y_1, \dots, Y_n$ and want to estimate π .

Step 1: Write down the likelihood

+ Let k be the number of times $Y_i = 1$ in the data

$$L(\pi) =$$

Maximum likelihood in more generality

Step 1: Write down the likelihood

$$L(\pi) = \pi^k (1 - \pi)^{n-k}$$

Step 2: Take the log

$$\log L(\pi) =$$

Maximum likelihood in more generality

Step 3: take the first derivative, and set = 0

$$\log L(\pi) = k \log(\pi) + (n - k) \log(1 - \pi)$$

$$\frac{d}{d\pi} \log L(\pi) =$$

Maximum likelihood in more generality

So our maximum likelihood estimate is $\hat{\pi} = \frac{k}{n}$, the sample proportion

+ Example data: T, T, T, T, H

+ This implies that $\hat{\pi} = \frac{1}{5} = 0.2$

Another way of viewing the problem

Suppose that $Y_i \sim \text{Bernoulli}(\pi)$. We observe n independent observations $Y_1, \dots, Y_n$ and want to estimate π .